

HOW TO USE A DIGITAL MULTIMETER – 10 min

Objective of Experiment No. 1

To become familiar with the use of the available digital multimeter by performing practical measurements of resistance, voltage (or potential difference), and current intensity.

Equipment

- 1 × Digital multimeter
 - 1 × Connector with red probe
 - 1 × Connector with black probe
 - 1 × AA battery (not included)
 - 1 × Inverted resistor box
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Description of the experiment

Voltage measurement:

Insert the red probe plug into the bottom right socket (labeled $V\Omega$) and the black probe plug into the COM socket. To measure the potential difference across a 5V battery, select the appropriate range so that it is not too large compared to the value to be measured (in this case 20V), and connect:

- red probe (+) to the positive terminal of the battery
- black probe (–) to the negative terminal of the battery

Current measurement:

Insert the black probe plug into the COM socket and the red probe plug into the second socket at the bottom left (labeled A). After selecting the appropriate range, connect the probes to the battery as in the previous measurement and observe the change in the displayed value.

Resistance measurement:

Insert the red probe plug into the bottom right socket (labeled $V\Omega$). Unlike current and voltage measurements, resistance can only be measured when the component is not powered.

This operation is performed by the multimeter by applying a small voltage, allowing it to evaluate the current flowing through the resistor, from which the resistance value is calculated.

If the component is powered, the measured resistance value will be incorrect.

In this case as well, first select the appropriate range and then place the multimeter probes on the two terminals of the resistor.

Questions for the teacher

1. Consider a 2.5V battery. Ignoring precision errors, what voltage value (V) would a multimeter display if set to a 2 mV range?
 2. Consider a 5V battery. What value would the multimeter display if the red probe is connected to the V Ω socket and the range is set to 20 mA?
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